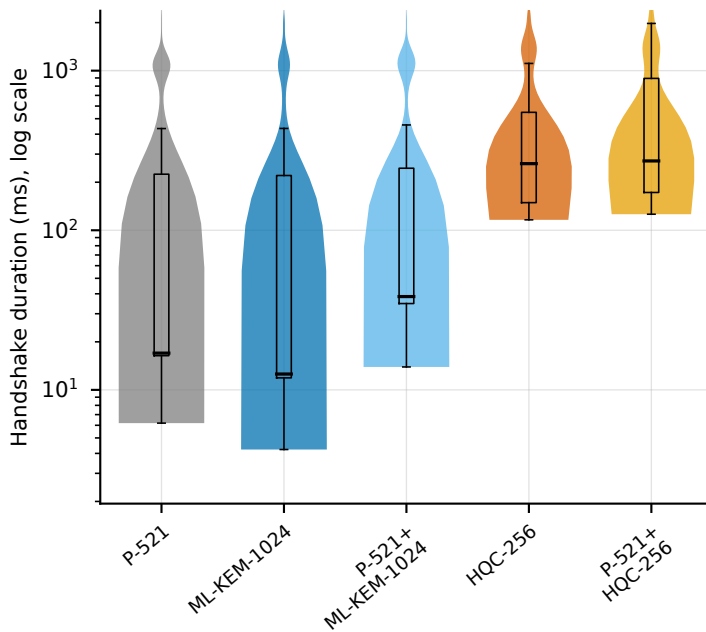
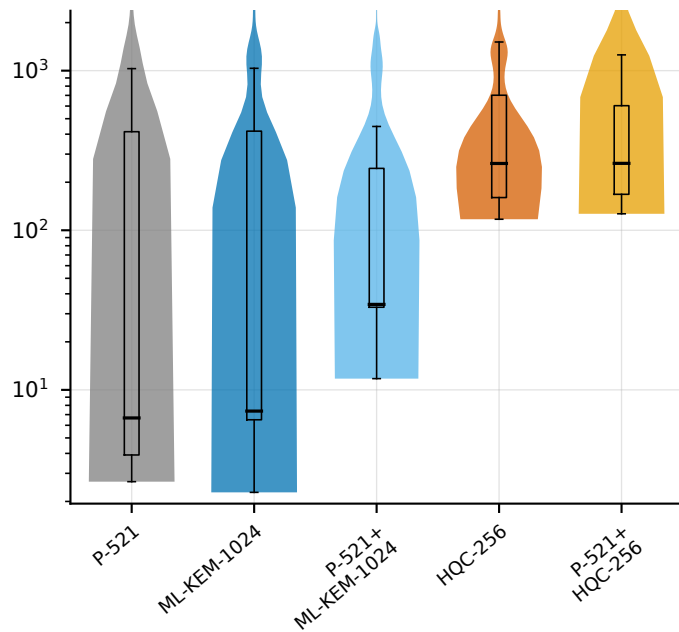


(a)

secp521r1 (classical)

**(b)**

mldsa87 (post-quantum)

**(c)**

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	secp521r1 Mean (ms)	secp521r1 CV (%)	secp521r1 Max (ms)	mldsa87 Mean (ms)	mldsa87 CV (%)	mldsa87 Max (ms)
P-521	307.2	189.9	5.15k	438.3	329.5	27.42k
ML-KEM-1024	286.4	214.3	5.12k	429.7	220.3	13.50k
P-521+ML-KEM-1024	328.2	191.6	6.41k	376.2	189.7	7.40k
HQC-256	569.0	134.8	6.78k	582.2	127.7	6.66k
P-521+HQC-256	666.4	135.4	7.89k	764.4	355.6	55.15k